

Question number	Answer	Additional guidance	Mark
3(a)(i)	A is the correct answer B is not the correct answer as a wider afferent arteriole does not increase water movement by osmosis into the Bowman's capsule C is not the correct answer as a wider afferent arteriole does not reduce the risk of capillary damage due to high pressure D is not the correct answer as a wider afferent arteriole does not reduce water reabsorption from the proximal tubule		(1)

Question number	Answer	Additional guidance	Mark
3(a)(ii)	C is the correct answer A is not the correct answer 6×24 does not equal 124 dm^3 B is not the correct answer as 6000×24 does not equal 144 cm^3 and it has not been converted to dm^3 D is not the correct answer as 6000×24 does not equal 1440 cm^3		(1)

Question number	Answer	Additional guidance	Mark
3(b)(i)	A calculation showing the following steps: • calculation of daily urine production (1) • answer given in standard form (1)	$1.5 \times 60 \times 24$ OR 2160 2.16×10^3 Accept 2.160×10^3 / 2.2×10^3 correct answer gains full marks	(2)

Question number	Answer	Additional guidance	Mark
3(b)(ii)	An answer that includes the following points: • as ADH (concentration) increases the urine concentration increases (1) • as ADH (concentration) increases the rate of urine production decreases (1)	ACCEPT 'it' for ADH concentration IGNORE higher urine concentration with lower urine production rate ACCEPT converse ACCEPT positive correlation between ADH concentration and urine concentration ACCEPT converse ACCEPT negative correlation between ADH concentration and rate of urine production	(2)

Question number	Answer	Additional guidance	Mark
3(c)	An explanation that includes three of the following points: • increases movement (of water) (1) • due to increased permeability of collecting duct (1) • (because ADH causes) {aquaporins / pores/ channels} to be inserted into {membrane / collecting duct} (1) • water moves {down water potential gradient /by osmosis} (1)	ACCEPT increases the reabsorption of water ACCEPT (ADH causes) increase in aquaporins in {membrane / collecting duct} ACCEPT high to low water potential	(3)

